Logic-Based Vending Machine Controller

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- **System Description:**

The system is a digital logic-based controller for a simplified vending machine designed to dispense a product priced at 15. The machine accepts two-coin denominations: 5 and 10. The logic circuit evaluates the inserted coins in real time to control two main operations: dispensing the product and returning change.

The product is dispensed automatically when the user inserts exactly 15 (one 10 coin and one 5 coin). If the user inserts two 5 coins and one 10 coin (totaling 20), the system satisfies the condition to dispense the item and simultaneously activates the 'Change' signal to return the extra 5.

Identification of Inputs and Outputs

Inputs:

Input A: Represents the insertion of a 5 coin.

Input B: Represents the insertion of a second 5 coin.

Input C: Represents the insertion of a 10 coin.

Outputs:

Out Signal: An output that triggers the dispensing mechanism when the required 15 is reached (represented by an LED indicator).

Change Signal: An output that activates the coin return mechanism to refund the excess 5 when 20 is inserted (represented visually using a Common Anode 7-Segment display).